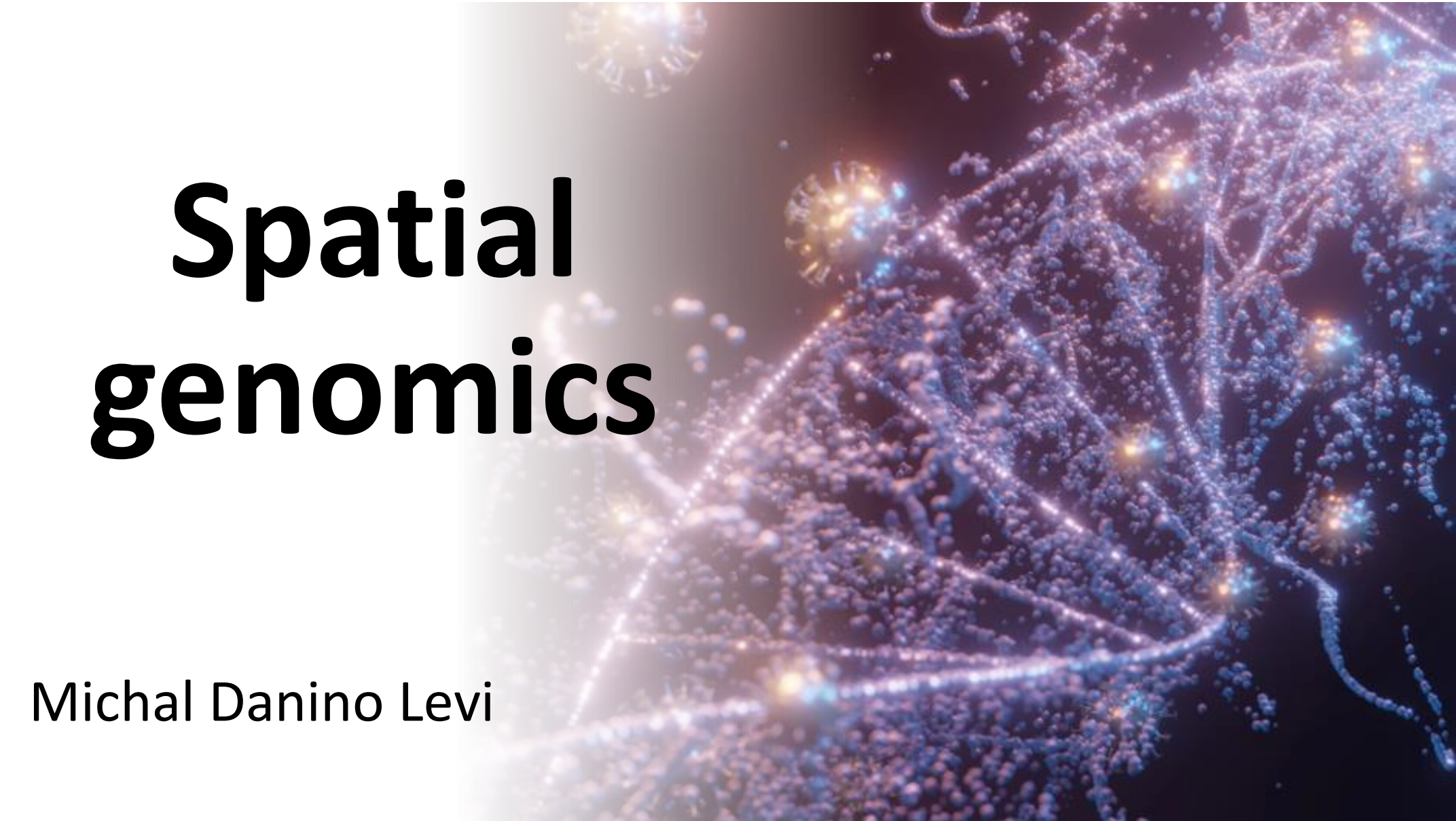


Spatial genomics

An abstract visualization of spatial genomics data. The background is a dark, deep blue space filled with a complex network of glowing blue and purple lines and dots, resembling a 3D map of genomic interactions or a neural network. Several bright, multi-colored starburst or nebula-like structures are scattered throughout, adding a sense of depth and complexity to the scene.

Michal Danino Levi

Bulk Sequencing



Single Cell Sequencing



Spatial Sequencing

	Tissue		Cell 1	Cell 2	Cell M		Cell 1	Cell 2	Cell M	X	Y	
Gene 1	108		2	8	17		2	8	17	21	86	
Gene 2	25		5	4	15		5	4	15	45	13	
Gene 3	19		9	35	3		9	35	3	66	91	
Gene 4	72		7	21	16		7	21	16	30	7	
Gene N	64		16	8	11		16	8	11	56	58	
Complex tissue							X	210	69	80	93	97
							Y	50	171	14	23	55
Cell isolation												
												

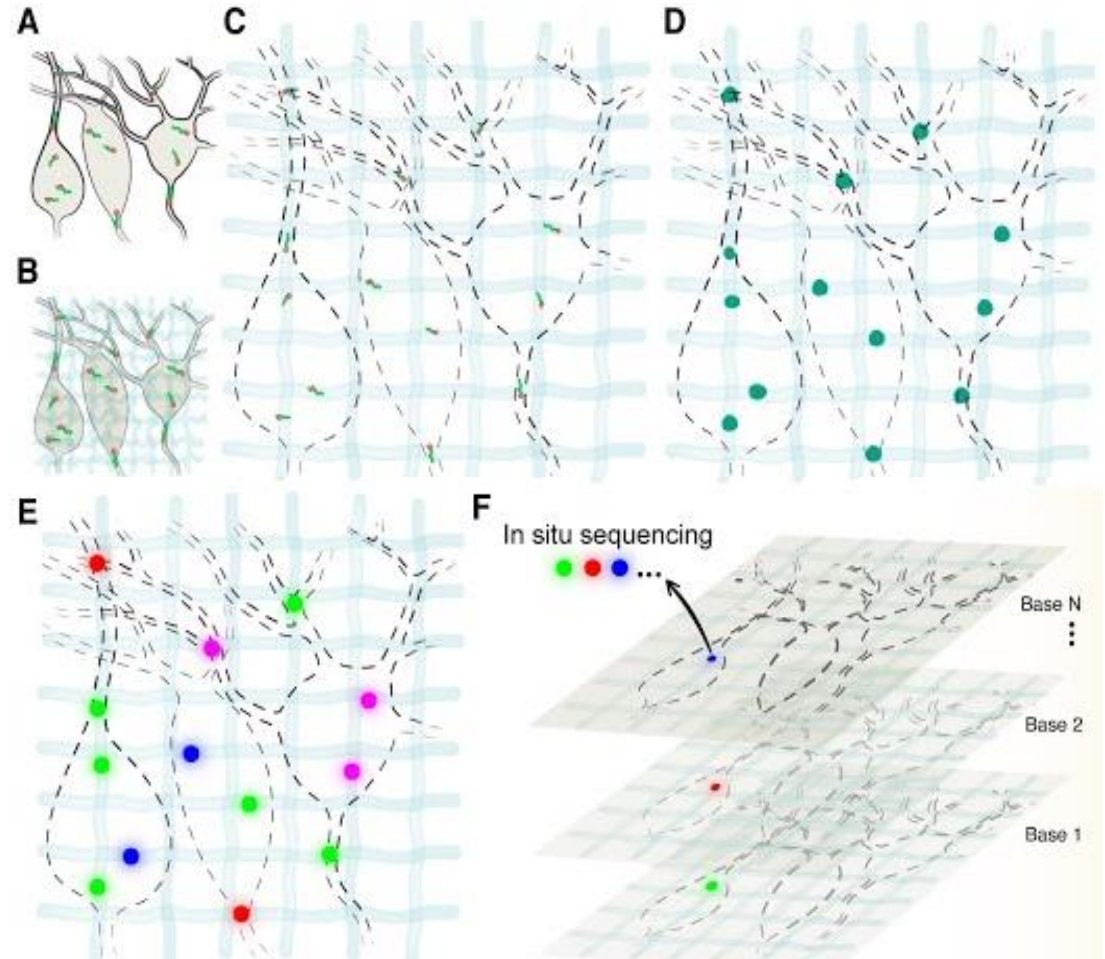


Expansion Sequencing - ExSeq

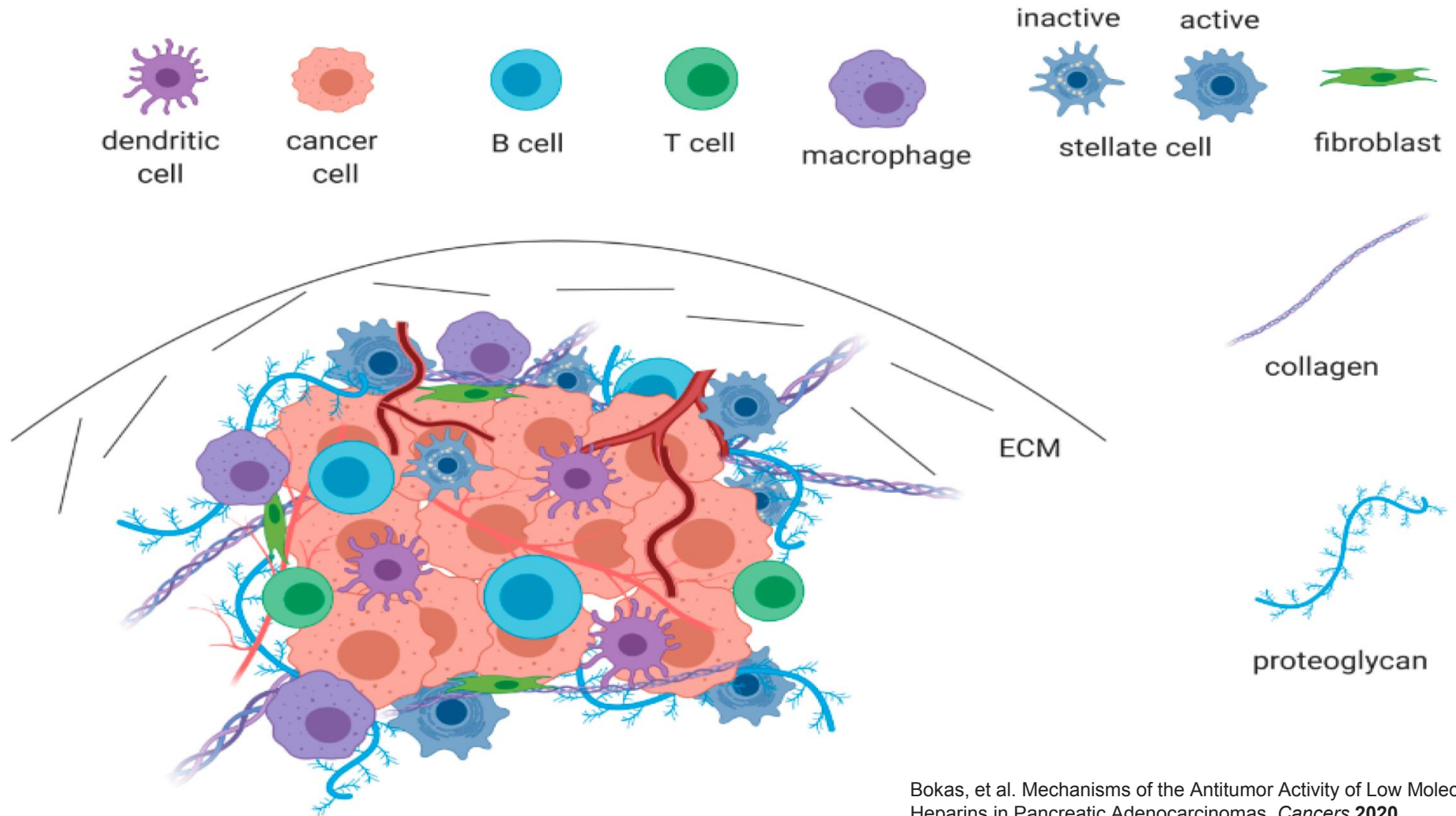
ExSeq allows the sequencing of hundreds of genes in their 3D original locations with nanoscale resolution.

Advantages:

- Number of genes
- Applicability to human samples
- Ability to analyze 3D samples
- Super resolution

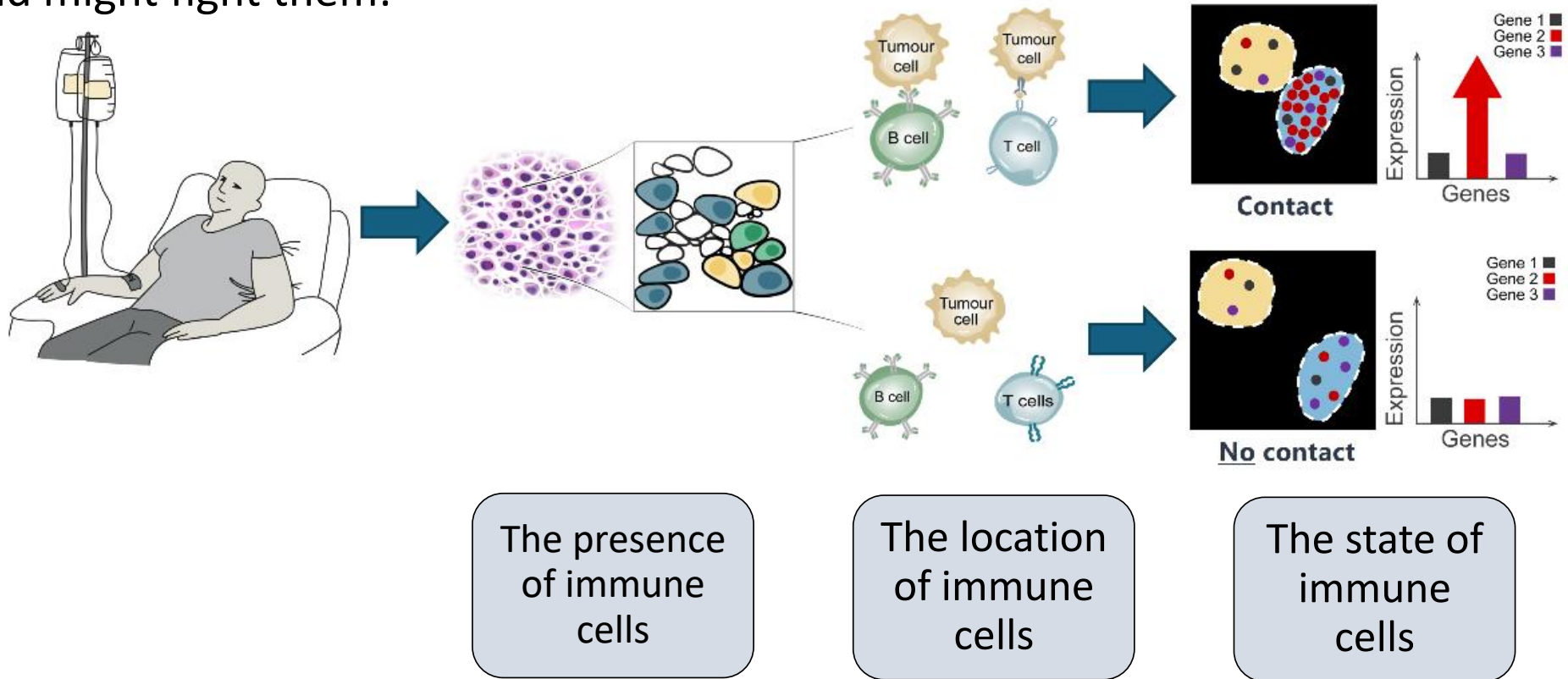


Tumor Microenvironment



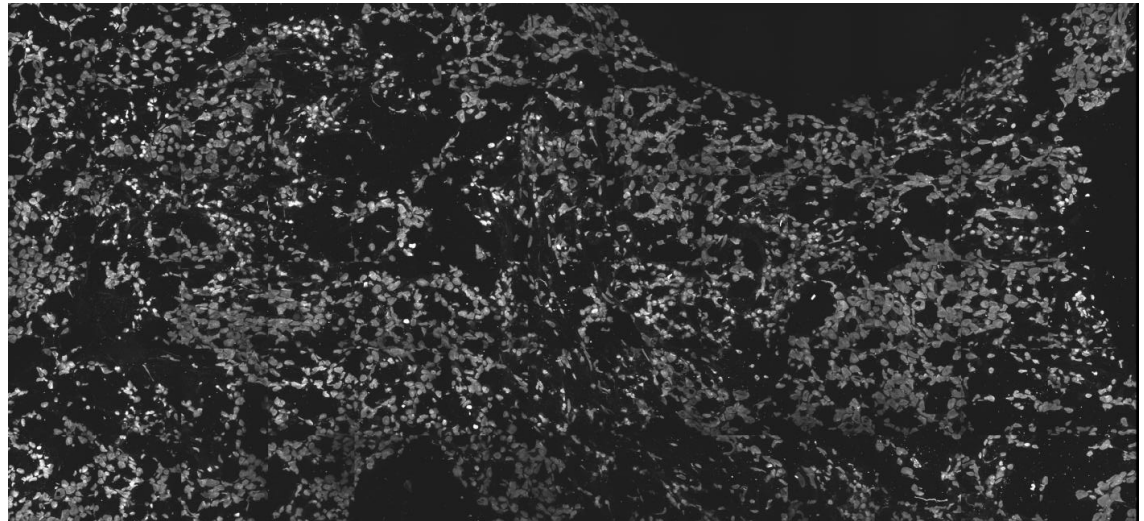
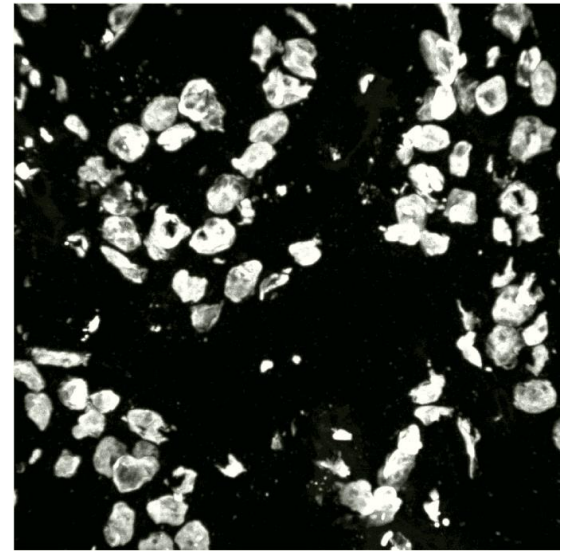
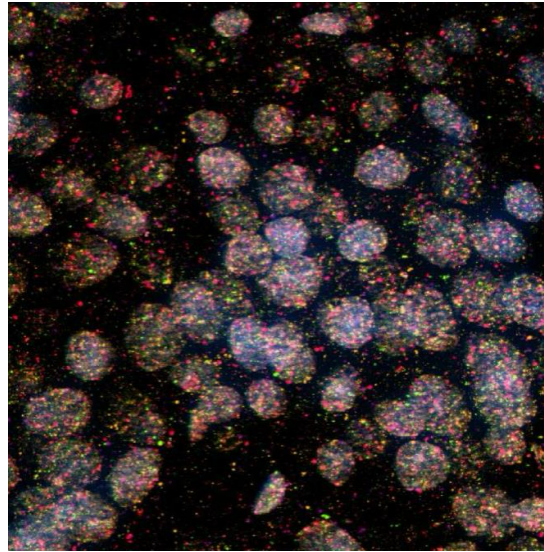
Characterization of Immune-Tumor Cells Interactions

At least three factors will determine if immune cells are responding to tumor cells, and might fight them:



Data

- Core biopsy taken from a patient with metastatic breast cancer infiltration into the liver.
- 297 genes.
- 1347 x 621 x 8 microns.
- DAPI staining of the nuclei.
- 76 fields of view of 40x objective.
- In situ sequencing via targeted ExSeq.
- Super resolution due to 3.3x physical expansion.

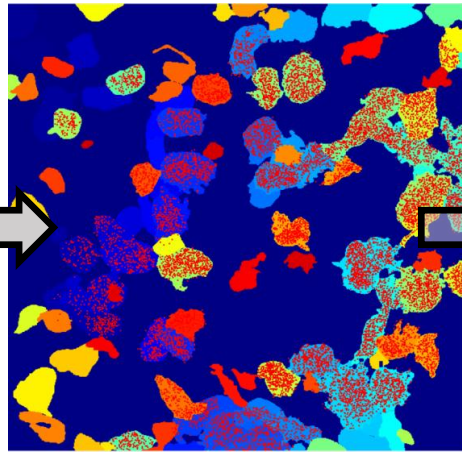
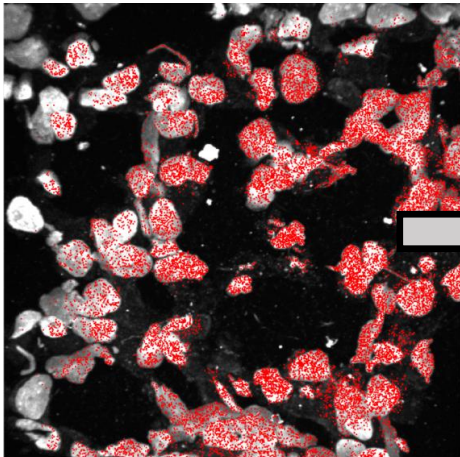


Research Plan

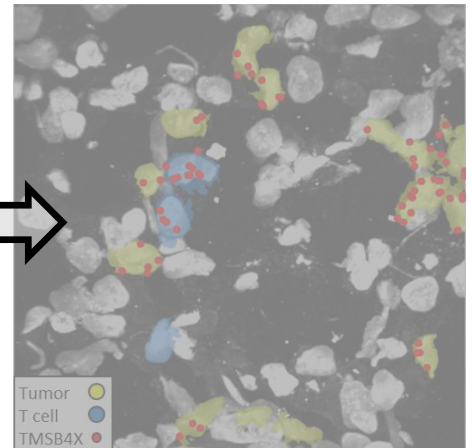
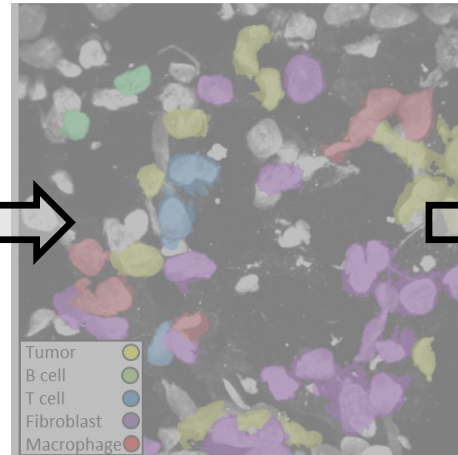


Identify genes that change their expression when immune cells are touching tumor cells

Create a 3D segmentation of single cells bodies in situ



Identify genes involved in spatial cell-cell interactions

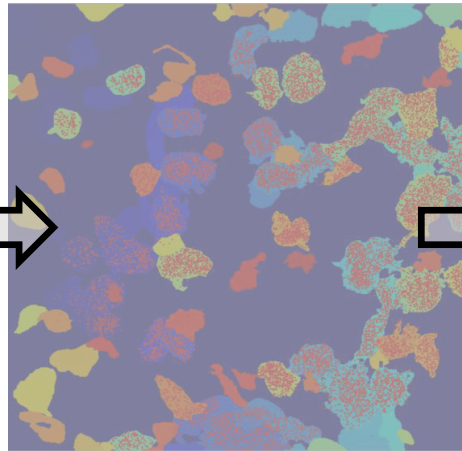
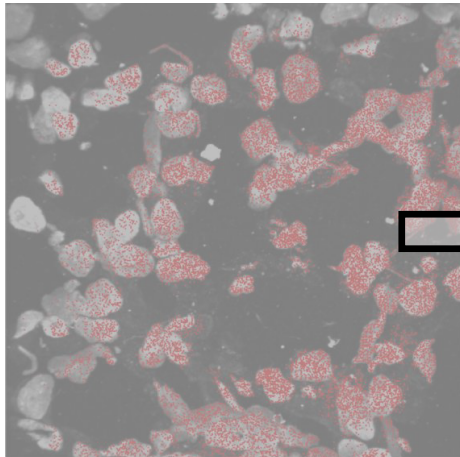


Research Plan

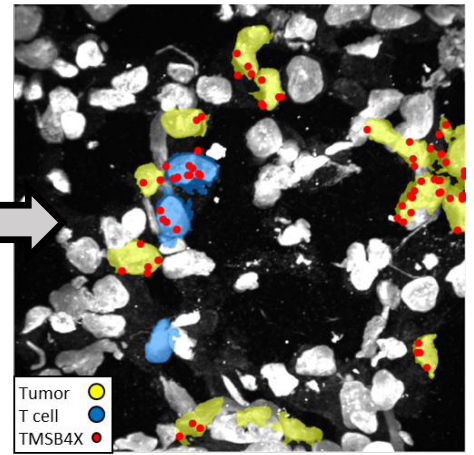
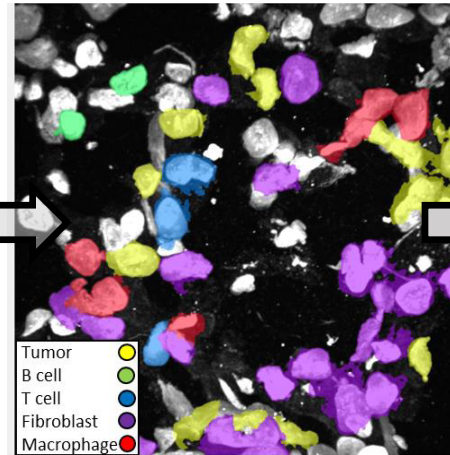


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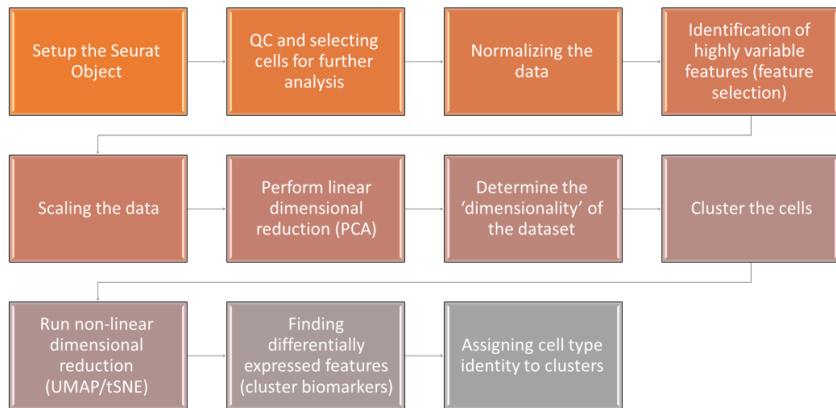


Identify genes involved in spatial cell-cell interactions

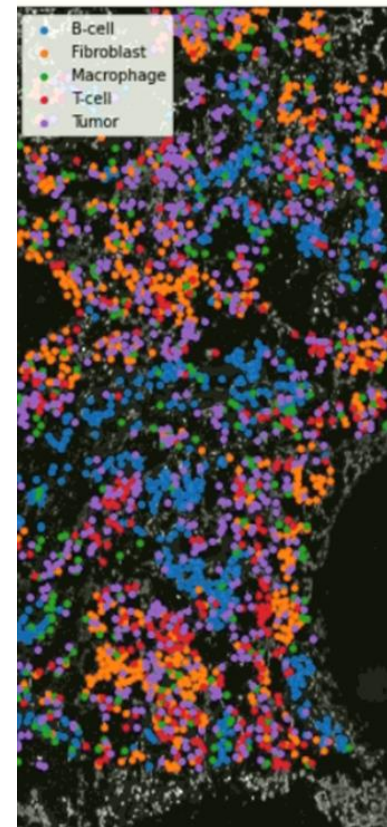
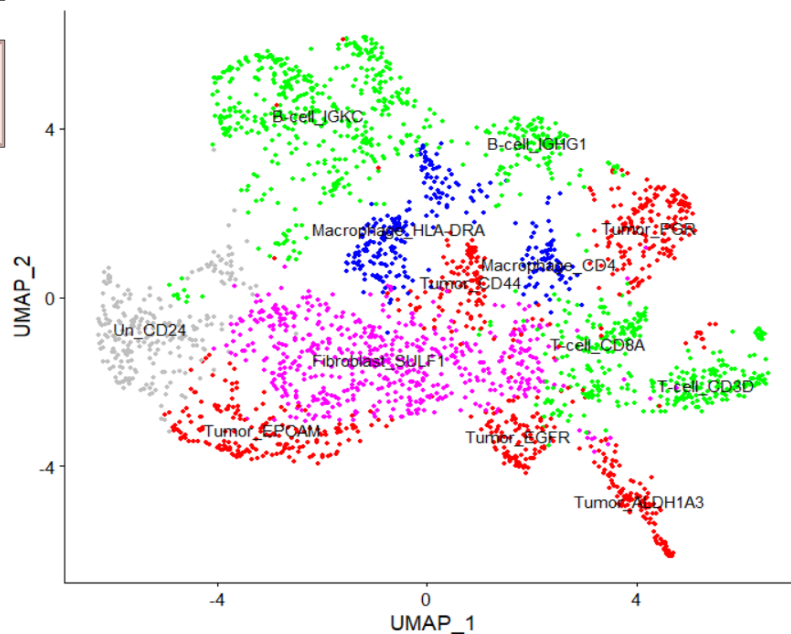
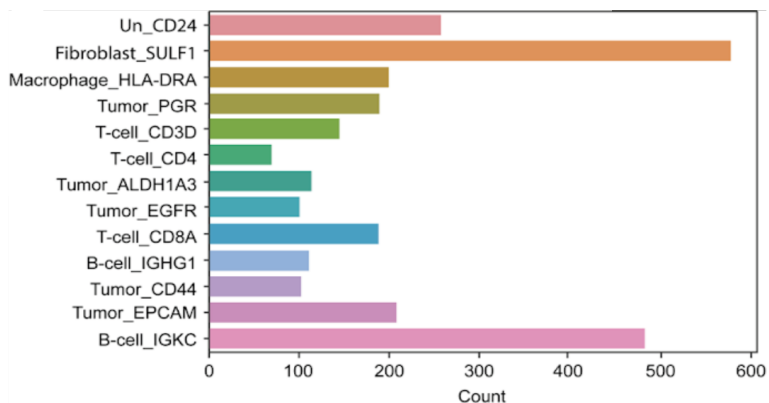


Clustering

UMAP representation of PCA-based expression clustering using Seurat



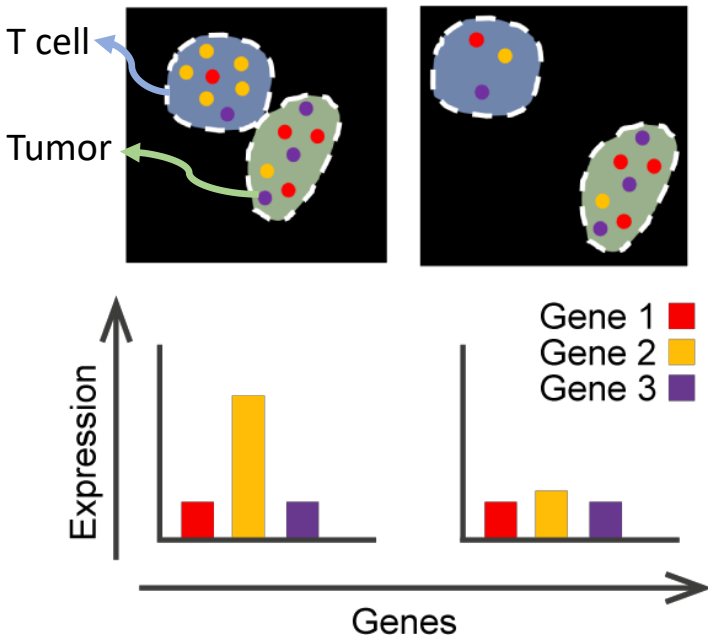
Seurat - Guided Clustering Tutorial – https://satijalab.org/seurat/articles/pbmc3k_tutorial.html



Genes Involved in Cell-Cell Interactions

Systematically detect genes, in specific cell types, whose expression separates immune cells that are proximal to tumor cells from those that are not proximal

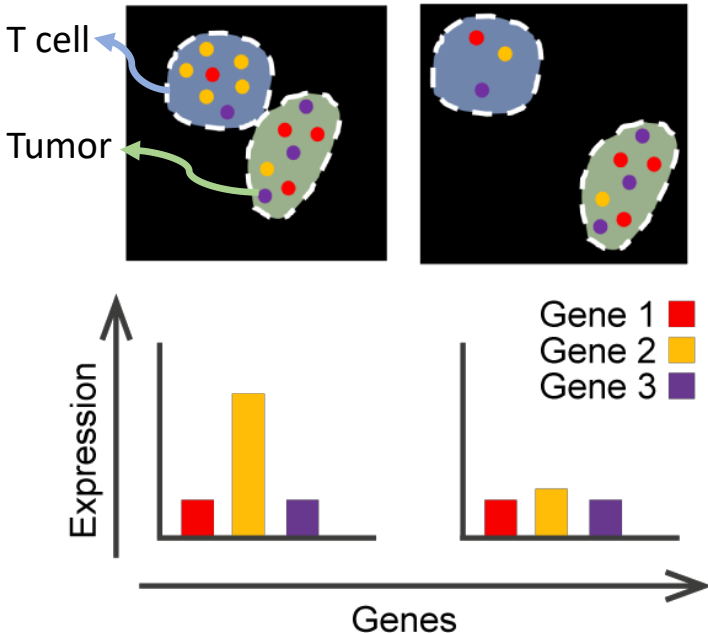
Differential Expression Analysis (DE)



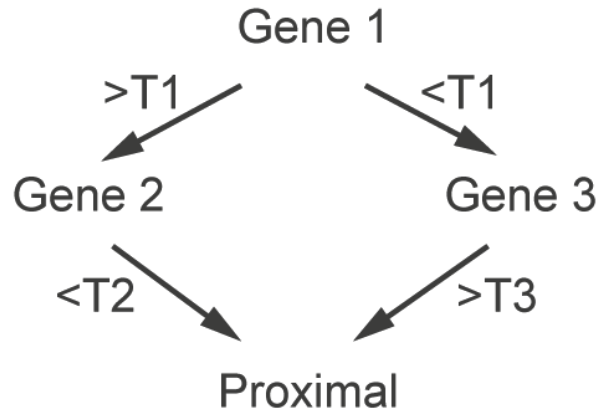
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Differential Expression Analysis (DE)



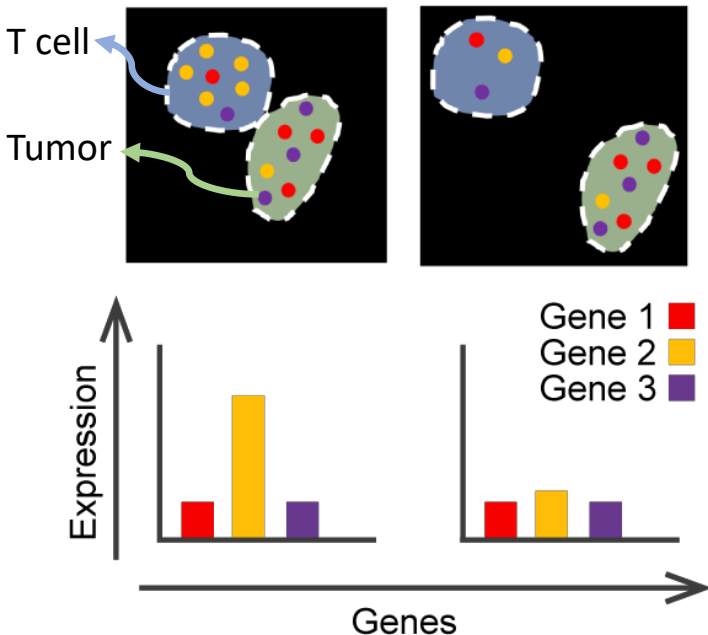
Machine Learning Classifiers (ML)



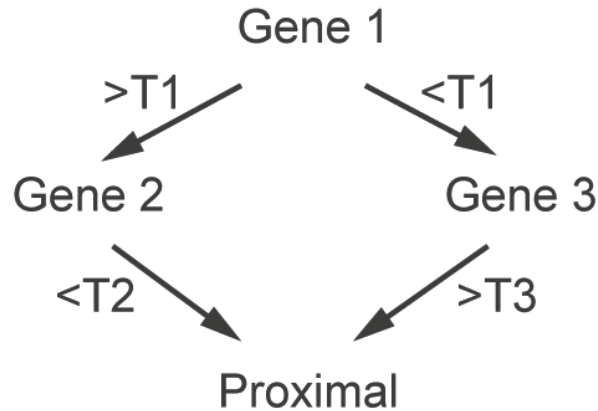
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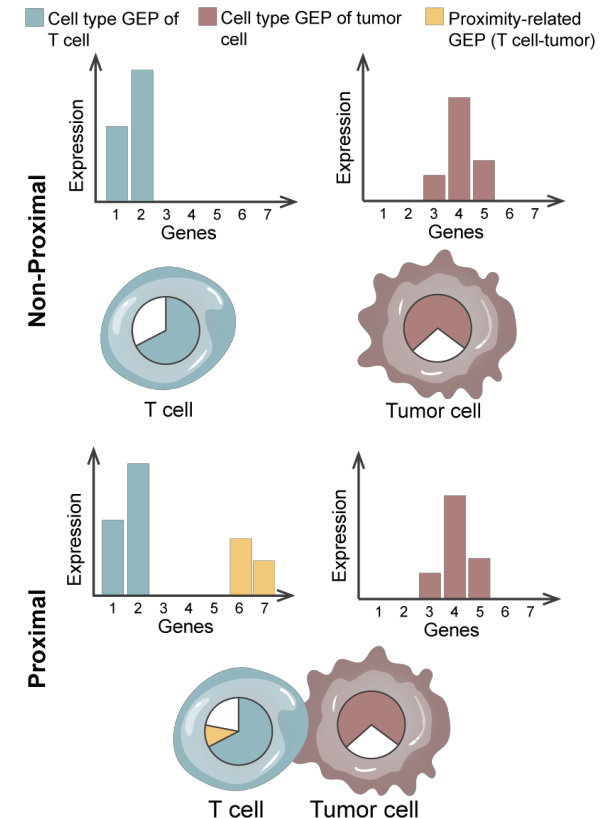
Differential Expression Analysis (DE)



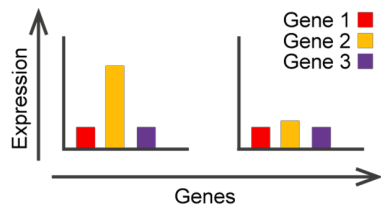
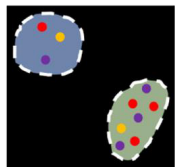
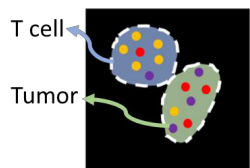
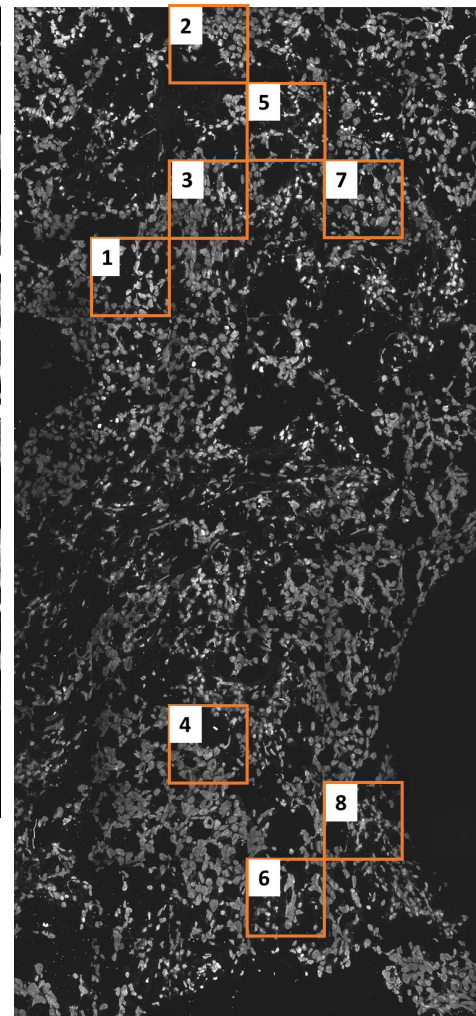
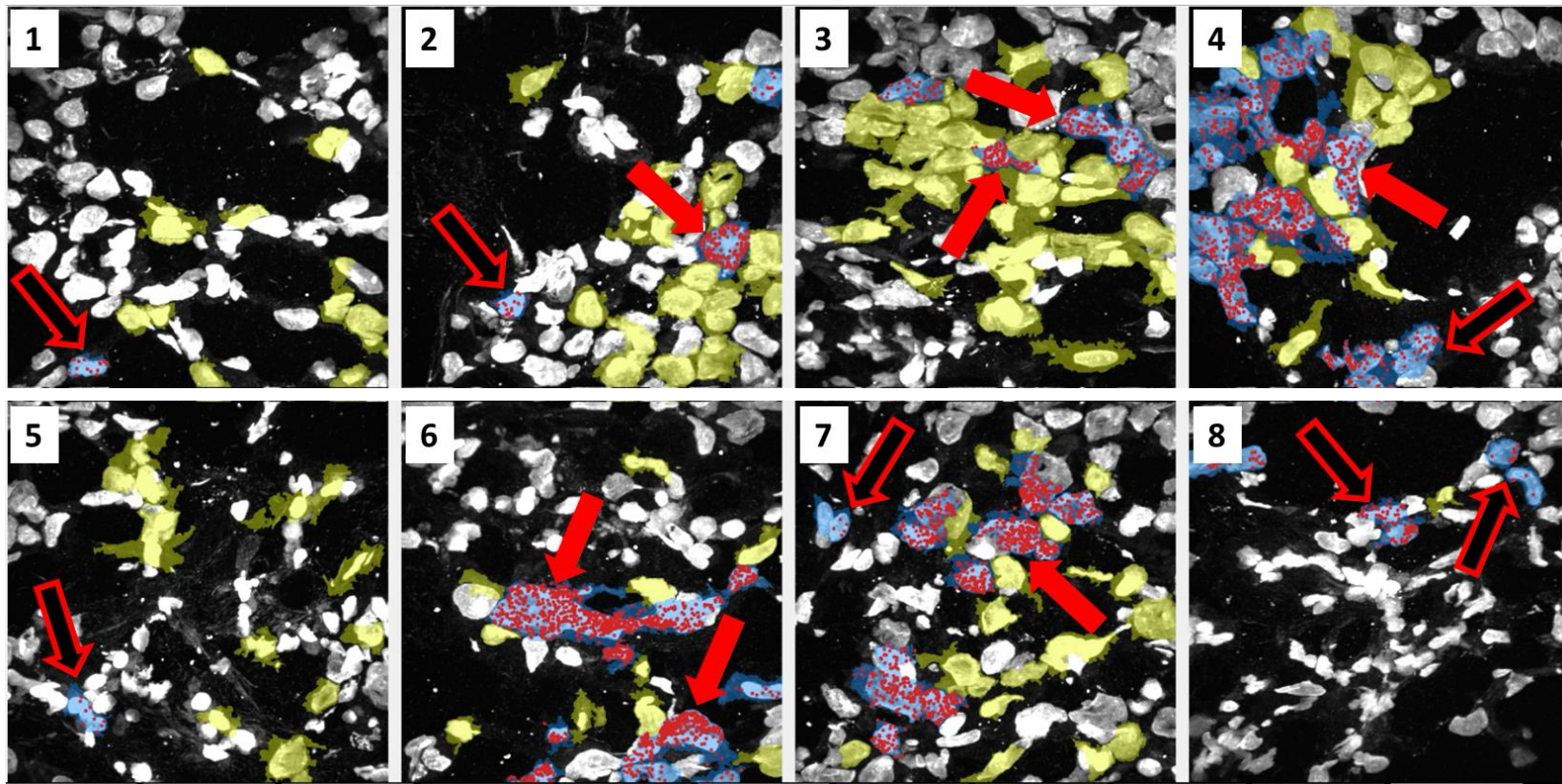
Machine Learning Classifiers (ML)



cNMF (MF)



Genes Involved in Spatial Cell-Cell Interactions



DE+ML methods